TAYLOR'S UNIVERSITY
Wisdom · Integrity · Excellence

SCHOOL OF COMPUTER SCIENCE

Group Project Assignment: 30%
Agile Software Solution for Sustainable Development

Declaration of Originality:

We hereby declare that the report entitled "_____" submitted for the **module ITS 610204/ITS 64604: PRINCIPLES OF SOFTWARE ENGINEERING, Semester: February 2026** is entirely our own original work; no part of this report has been copied from other students, published works, or any other source without proper acknowledgment, and no unauthorized assistance (including any AI text generation tool) has been used in preparing this report, and we understand that any violation of this declaration may result in academic penalties.

Student Name	**Student ID**	**Signature**
Group Leader Name:		
Member:		
Member:		
Member:		
Member:		

MLO3: Demonstrate subject specific skills with respect to apply modelling techniques and notations in the group project coursework.

Syllabus covered: Introduction to Software Engineering, Software Development Processes, Software Tools and Environments, Agile Software Development, SCRUM tools.

Assigned Date: Week 2, 11th February 2026 **Due Date:** Week 6, 13th March 2026, 5:00PM

1. Project Overview

This 4-week group assignment focuses on Software Engineering Tools, Agile and Scrum methodology, and teamwork skills. Each group of 5 must develop a prototype for a software-based solution related to ONE of the following Sustainable Development Goal (SDG).

A. **Quality Education (SDG 4):**
Software can improve learning access and skill development.
Example: Online tutoring platform for high school students.

B. **Good Health and Well-Being (SDG 3):**
Software can help track health and provide timely medical support.
Example: Appointment booking system for a university clinic.

C. **Clean Water and Sanitation (SDG 6):**
Software can monitor water quality and manage community resources.
Example: Mobile app for reporting and tracking local water issues.

D. **Sustainable Cities and Communities (SDG 11):**
Software can optimize urban living and community engagement.
Example: Smart waste management tracking platform.

E. **Climate Action (SDG 13):**
Software can raise awareness and track environmental impact.
Example: Carbon footprint tracking app for students.

2. AI Usage Requirements:

Students must be cautious and ethical when using AI tools. AI MAY BE used for inspiration or general research ONLY, but NOT for producing any work submitted for grading. The following rules apply:

- AI may NOT generate any part of the report, prototype, backlog, diagrams, user stories, or slides.
- AI may NOT design UI/UX, generate images, or produce Agile artifacts.
- AI essays, diagrams, or code are STRICTLY PROHIBITED.
- AI may be used only to clarify concepts (e.g., 'What is a burndown chart?').
- All work must be original, manually created by the group.
- Students are ENCOURAGED to use proper Academic Citation as and when applicable.

<u>WARNING:</u> *If students use AI to generate any deliverable content, that section will receive ZERO MARKS. This project is designed to evaluate collaboration, problem-solving, software engineering tools, and Agile/Scrum understanding, not AI prompting output.*

3. Project Deliverables

- Product Vision Document including chosen SDG and justification
- Requirements Gathered & Documented by using appropriate Modelling techniques & tools for Use Case Diagram
- Agile Artifacts: Product Backlog, Sprint Backlogs, Burndown Charts, Kanban Board (Please Use Trello and demonstrate your use in presentation, and report it appropriately)
- Trello Board Activity Evidence (weekly snapshots)
- Working SDG-related Prototype
- GitHub Repository Activity Evidence (Provide meaningful commit messages and include screenshots or extracts as evidence of progress in planning, SDLC work, and alignment with Trello tasks)
- Final Group Presentation (Marks allocated based on individual performance)

Each group must use both **GitHub** and **Trello** as follows:

a) GitHub Requirements:

1. Create Your Team GitHub Repository
2. Write a Clear README File that includes:
   - Project Title & SDG Chosen : which Sustainable Development Goal your app prototype supports.
   - Team Members : names and roles.
   - Project Description : what your idea is and who it helps.
   - Overview of Your SDLC Work : short summary of how you went through planning, design, testing/evaluation.

This helps teachers and reviewers understand your project at a glance.

3. Organize Your Files in Folders, store your work logically inside folders, such as:
   - `/planning/` : Project plan, Trello links, requirements, sprint notes
   - `/design/` : Wireframes, mockups, related diagrams
   - `/testing/`: Prototype feedback, notes on user testing
   - `/final/` : Final prototype screenshots, summary of results

This organization shows the SDLC phases clearly and makes your work easy to review.

4. Commit Your Work Regularly, even if you are not writing code:

   ✓ Upload new files or updates often.

   ✓ Write short but clear commit messages like "Add initial user stories", "Upload wireframes for UI design", "Add prototype evaluation feedback".

These commits become a timeline of your project progress (like a diary of your work).

<u>Proof required:</u> Screenshots or extracts from your GitHub repository showing the repository setup, README file, folder organization, commit history (with meaningful messages), and any issues or linked tasks that demonstrate your project progress and SDLC work from start to finish.

b) Trello Requirements:

- Create and break down user stories / tasks / issues

- Track progress using boards (Scrum)
- Tasks represented by cards for all SDLC stages.
- Show movement of cards/tasks through stages (e.g: `To Do → Doing → Done`)
- Assign tasks to team members, show deadlines and completion dates
- Produce visual project timelines, sprint boards, and burndown charts (if supported)
- Backlog tasks aligned with Product Backlog and Sprint Backlogs.

Proof required: Screenshots/extracts showing board setup and task flow (backlog, sprints, labels, assignments) from start to finish.

4. Weekly Work Plan (4 Weeks)

Week 1 – Initiation & Planning:
Form group, assign roles, choose SDG, create Product Vision, initial backlog, set up Kanban board.

Week 2 – Sprint 1 (Design & Early Prototype):
Sprint planning, UI sketches, early prototype, test plan, Sprint Review & Retrospective.

Week 3 – Sprint 2 (Build & Test):
Develop improved prototype, run tests, bug reports, updated burndown chart, Sprint Review & Retrospective.

Week 4 – Finalization & Presentation:
Finalize prototype, prepare slides, rehearse, compile full report, final submission. Your presentation must be interesting & engaging. Excellent student will be able to present like a Sale Pitch for the prototype developed.

6. Report Template*

1. Title Page / Cover Page
2. Abstract
3. Project Introduction
4. Agile Process / Trello Usage
5. SDLC Phases
    - Planning / Requirements
    - Analysis
    - Design
    - Prototype Development
    - Testing / Feedback
    - (Optional) Deployment Summary
6. GitHub Evidence
7. Prototype Showcase
8. Reflections & Lessons Learned
9. Conclusion

Appendices (Screenshots / Evidence)

*This is a report Template guide only. Students may have additional sections as and when required, provide justifications for each additional sections included.

*** Please Include this Rubrics in your Report Submission*

Project Report Rubric (20 Marks)

SDLC Phase	Performance Level	Description	Marks Awarded (Lecturer please circle the marks awarded)	Total Marks / Comments
Requirements & Planning (SDLC – Planning & Analysis)	Outstanding (2 marks)	Requirements and planning are **clear, well-organised, and aligned** with project goals and SDG focus, and there is **strong evidence in Trello and GitHub** (user stories, backlog items, and planning files).	2	
	Average (1 marks)	Requirements and planning are **present but somewhat incomplete or unclear**, with some supporting evidence in Trello/GitHub but lacking in detail or organisation.	1	
	Poor (0 mark)	Requirements and planning are **unclear, missing, or poorly aligned** with project goals/SDG focus, with **little or no evidence** in Trello or GitHub.	0	
Use Case Diagram (SDLC – Requirements Analysis)	Outstanding (4 marks)	A **Use Case Diagram is submitted** that correctly follows UML notation (actors, use cases, system boundary, and relationships) and effectively represents the main functional interactions of the system.	4	
	Average (2-3 marks)	Use Case Diagram is present but **has one or more notation errors** or missing elements (e.g., missing actors, relationships, or unclear boundaries), though basic interactions are shown.	2 / 3	

	Poor (0 -1 mark)	Use Case Diagram is **missing or does not follow UML notation**, and fails to present functional interactions clearly.	0 / 1	
Design & Prototype Development (SDLC – Design & Implementation)	Outstanding (4 marks)	Design artifacts (wireframes, mockups or prototype screens) are **clear, logical, and user-focused**, the prototype description clearly explains features and value, and the **GitHub design folder is well organised** with **meaningful commit history** showing design progress over time.	4	
	Average (2–3 marks)	Design artifacts and prototype are present and understandable but lack clarity or completeness, and the GitHub design folder contains some organisation and commits but not fully clear or comprehensive.	2 / 3	
	Poor (0–1 mark)	Design artifacts or prototype **are missing, unclear, or not linked to user value**, and the **GitHub design folder lacks proper organisation or meaningful commits**.	1 / 0	
Agile Process & Tool Usage	Outstanding (4 marks)	Able to **clearly explain** how Agile practices (e.g., iterative planning, task prioritization, feedback/reflection loops) were applied and **explicitly describe how Trello (and other tools, if used) supported these practices**, with specific examples of boards, lists, cards, and updates that show real workflow and adaptation.	4	
	Average (2–3 marks)	Able to **describe some Agile practices** and mention use of Trello, but the explanation is **general or incomplete**, with limited specific examples of how the tool supported workflow or decisions.	2 / 3	

	Poor (0–1 mark)	**Unable to explain** Agile practices meaningfully, or the explanation does not include how Trello was used to support the Agile process in the project.	1 / 0	
Testing / Evaluation (SDLC – Testing)	Outstanding (4 marks)	Able to clearly explain a structured evaluation with **defined test goals and tasks**, include **user-focused feedback or usability testing results**, and show how the findings were used to improve the prototype.	4	
	Average (2–3 marks)	Able to describe some evaluation activity with **basic explanation of tasks or feedback**, but the process or connection to improvements is **limited or only somewhat systematic**.	2 / 3	
	Poor (0–1 mark)	Unable to explain meaningful evaluation or testing activity, **lacking clear goals, tasks, feedback, or evidence of improvements** based on evaluation.	1 / 0	
Reflection & Professionalism (SDLC – Close & Reflect)	Outstanding (2 marks)	Able to clearly explain how team members collaborated and used tools (Trello, GitHub) to support their work, describe what was learned about SDLC and Agile, and present artefacts and documentation in a clear, organised, and professional way.	2	
	Average (1 marks)	Able to describe some aspects of team collaboration and tool usage with some learning about SDLC/Agile, and documentation is present but only somewhat clear or moderately organised.	1	
	Poor (0 mark)	Unable to meaningfully explain team collaboration or tool usage, with little reflection on SDLC/Agile learning and	0	

		documentation that is unclear, disorganised, or incomplete.		
Total Marks (Project Report)				_____/ 20

Individual Presentation Rubric (10 Marks)

This rubric evaluates individual understanding, communication, and professionalism.

SDLC Phase	Performance Level	Description	Marks Awarded (Lecturer please circle the marks awarded)	Total Marks / Comments
Understanding of Project & SDG	Outstanding (4 marks)	Clear and deep understanding of SDG, software solution, personal contribution, and Agile/Scrum processes.	4	
	Average (2–3 marks)	General understanding with some gaps; basic Agile/Scrum knowledge.	2 / 3	
	Poor (0–1 mark)	Misunderstands SDG; unclear role explanation; little/no Agile/Scrum understanding.	1 / 0	
Communication & Delivery	Outstanding (4 marks)	Clear, confident, organized speech with strong pacing.	4	
	Average (2–3 marks)	Understandable but with pauses or unclear moments; moderate confidence.	2 / 3	
	Poor (0–1 mark)	Hard to understand; disorganized; very low confidence; poor timing.	1 / 0	
Visuals & Professionalism	Outstanding (2 marks)	Well-organised, non-AI slides; evidence of SE tools; professional delivery.	2	
	Average (1 marks)	Clear slides; minimal tool evidence; moderately professional.	1	
	Poor (0 mark)	Slides messy or AI-generated; lacks tool evidence; unprofessional delivery.	0	

Total Individual	_____ / 10
Total Marks (Project Report + Individual Presentation)	_____ / 30